

User Manual – LIDA Composting System

1. Introduction

The LIDA Composting System is a 3D-engineered composting unit that integrates hardware and software components to simulate and monitor composting behaviour. The model demonstrates airflow, gas exhaust, and moisture drainage systems, supported by sensor feedback (gas, temperature, and moisture). A companion mobile application visualises sensor data and allows user interaction with the composting model.

2. System Overview

The system comprises:

- 3D Model (SolidWorks/SketchUp): Represents the composting chambers, pipeline system, paddles, and drainage tanks.
- Sensor System: Gas, moisture, and temperature sensors simulate real-time data collection.
- Motor System: Controls paddle rotation and airflow for compost agitation.
- Mobile Application (Flutter-based): Displays live sensor data, composting stages, and animations.
- User Interface (UI): Provides real-time feedback and visual cues through a monitor integrated into the model.

3. User Requirements

1. A compatible device (Windows PC or laptop) capable of running the 3D simulation.
2. Access to the Flutter mobile app for sensor data display.
3. Stable internet connection for app synchronization (if using cloud data simulation).

4. Getting Started

- Step 1: Launch the LIDA 3D model file (.SLDASM or .GLB format).
- Step 2: Use the control panel or animation manager to play composting sequences (paddle rotation, airflow, and condensation).
- Step 3: Open the LIDA Mobile App to view simulated sensor readings.
- Step 4: Ensure that all components (sensors, water tank, and monitor) are correctly loaded and visible before simulation.

5. Using the Mobile Application

1. Home Screen: Displays overall composting status and stage indicators (Active, Curing, Finished).

Sensor Data Screen: Shows real-time simulated readings for:

- Temperature (°C)
- Gas concentration (CO₂/CH₄)
- Moisture percentage (%)

2. Controls:

- Toggle airflow or paddle rotation.
- Refresh data or trigger manual updates.
- Monitor composting chamber through live visuals.

6. Maintenance and Troubleshooting

Issue	Cause	Solution
Sensor readings not displaying	App not synced or access permissions not granted	Reconnect to mobile app or reload simulation
Animation delay	Model too complex / hardware lag	Reduce animation layers or simplify textures
Drainage not visible	Hidden components	Enable transparent mode in 3D viewer
Incorrect compost colours	Material layer conflict	Reassign colour codes in SolidWorks

7. Safety and Ethical Guidelines

- All simulated data is for educational demonstration only; it does not represent real-world composting metrics.
- Ensure compliance with Australian Privacy Principles (APPs) when handling or sharing any sensor data.
- Avoid editing model components without approval to prevent system instability.

8. Contact and Support

For assistance or queries related to the system setup, contact the LIDA Project Team (Group 2):

- Thenura Kuruppuarachchi – Integration & Coordination
- Haritha Perera – Animation & Simulation
- Senupama Deshapriya – Research & Piping Design
- Chrishelle Anthoneyz – Structural Design & Documentation